Phân công:

- 2 topics (Association Rules Mining, Recommender System): Section 1 (mỗi topic 10 câu) + Section 2 (mỗi topic 5 câu) + Section 3: Thảo
- 3 topics (Fuzzy Logic, Linear Regression, Logistic Regression): Section 1 (mỗi topic 10 câu) + Section 2 (mỗi topic 5 câu): Nguyên
- 3 topics (Latent Dirichlet Allocation, Deep Neural Networks, Word Embedding): Section 1 (mỗi topic 10 câu) + Section 2 (mỗi topic 5 câu): Lịch

POST-TEST

SECTION 1:

Association Rules Mining

No.	Bloom's Level	Question
1	Remembering	**According to the formal definition of an association rule X -> Y, which condition must be met regarding the relationship between set X and set Y?** A. X must be a subset of Y (X is a subset of Y) B. X and Y must be disjoint (X intersection Y = empty set) C. X and Y must contain the same number of items D. X union Y must equal the set of all items I
2	Remembering	**In the apriori_gen procedure, what is the specific condition for joining two (k-1)-itemsets (L1 and L2) to create a candidate k-itemset?** A. They must not share any common items B. Only their first items must be identical C. Their first (k-2) items must be identical, and L1[k-1] < L2[k-1] D. All items in L1 must be smaller than all items in L2
3	Understanding	**Why does the Apriori algorithm perform a "Prune step" by checking if all (k-1)-subsets of a candidate are frequent?** A. To ensure that the candidate has a higher support than its subsets B. Because if any subset is infrequent, the superset cannot possibly be frequent C. To reduce the memory usage by deleting the database D. To allow the algorithm to generate rules with low confidence

4	Understanding	**The ARM problem is decomposed into two sub-problems. What is the primary focus of the first sub-problem?** A. Calculating the Lift and Conviction of all rules B. Generating rules from every possible item combination C. Finding all itemsets whose support count is >= minSup D. Scanning the database only once to find the maximum transaction length
5	Applying	**Given L2 = { {I1, I2}, {I1, I3}, {I1, I5}, {I2, I3} }. During the generation of C3, which candidate is generated by the Join step but then Pruned?** A. {I1, I2, I3} B. {I1, I2, I5} C. {I1, I3, I5} D. Both B and C
6	Applying	**Based on the transaction table (T100-T900), calculate the Confidence of the rule {I1, I3} -> {I2}.** A. 33.3% B. 50% C. 66.7% D. 75%
7	Analyzing	**If the minimum support (minSup) is significantly decreased, how does it affect the computational complexity of the Apriori algorithm?** A. Complexity decreases because fewer candidates are generated B. Complexity increases because more itemsets will satisfy the threshold, leading to more iterations and database scans C. It has no effect on complexity, only on the number of rules D. The algorithm stops after the first scan of the database

8	Analyzing	**Compare the Support of a rule X -> Y and the Support of its antecedent itemset X. Which statement is always true?** A. Support(X -> Y) >= Support(X) B. Support(X -> Y) <= Support(X) C. They are always equal D. They are only equal if the confidence is 50%
9	Evaluating	**A retail manager discovers a rule with high Confidence but very low Support. How should they evaluate this result?** A. It is a highly reliable rule that applies to the entire customer base B. It is a "rare rule" that is very strong for a specific small niche but lacks statistical significance for the whole store C. It is a data error because Support must always be higher than Confidence D. They should increase minSup to find more rules like this
10	Creating	**How would you modify the has_infrequent_subset procedure if you wanted to find "Closed Itemsets" (itemsets where no superset has the same support)?** A. By deleting candidates that have the same support as their subsets B. By skipping the join step entirely C. By adding a check to see if any superset S (where S contains C) has Support(S) = Support(C) D. By only returning itemsets that contain exactly 2 items

Recommender System

No.	Bloom's Level	Question

1	**Remembering**	Which accuracy metric is specifically noted for placing a higher emphasis on larger deviations between predicted and actual ratings? A. Mean Absolute Error (MAE) B. Root Mean Square Error (RMSE) C. Precision at k D. Recall
2	**Remembering**	In the context of Content-based Filtering, what does the Inverse Document Frequency (IDF) aim to achieve? A. Measuring the frequency of a term within a single document B. Reducing the weight of terms that appear in almost all documents C. Normalizing the document length to avoid bias D. Identifying the semantic meaning of keywords
3	**Understanding**	Why is Item-based Collaborative Filtering (I2I) often preferred over User-based (U2U) in large-scale commercial systems like Amazon? A. User similarities are more stable than item similarities B. It eliminates the cold start problem for new users C. Item relationships are more stable and the number of items is often smaller than users in such domains D. It requires no pre-processing of data
4	**Understanding**	From the "Recommendation Perspective," what is the primary goal of suggesting items from the "Long Tail"? A. To recommend the top 20% most popular items B. To identify widely unknown items that users might actually like (Serendipity) C. To reduce search costs for items the user already knows D. To optimize for the highest possible Precision score
5	**Applying**	If a keyword i appears in every single document in a collection ($n(i) = N$), what will its TF-IDF weight be regardless of its frequency in a document? A. 0 B. 1 C. ∞ D. TF(i,j)

6	**Applying**	In User-based CF, when calculating a prediction for Alice, why do we add the weighted difference of neighbors' ratings to **Alice's own average rating** $\overline{r}_a$? A. To simplify the mathematical calculation B. To account for differences in individual rating behavior (bias) C. To ensure the final rating is between 1 and 5 D. To prioritize items that Alice has already seen
7	**Analyzing**	What is the fundamental disadvantage of using the "Memory-based" approach in a real-world system with tens of millions of customers? A. It requires excessive knowledge engineering effort B. It cannot handle explicit user feedback C. It does not scale because the entire rating matrix must be used at run-time D. It is less accurate than simple Content-based filtering
8	**Analyzing**	How does "Recursive Collaborative Filtering" address the challenge of sparse datasets? A. By switching to a non-personalized recommendation strategy B. By increasing the minimum threshold for co-rated items C. By predicting ratings for neighbors who haven't rated an item yet to use as input for the active user D. By only recommending items from the "Head" of the popularity curve
9	**Evaluating**	A researcher finds that a system has very high RMSE but excellent user satisfaction scores. Based on the "Interaction Perspective," why might this happen? A. Accuracy metrics like RMSE are the only true measure of success B. The system may excel at explaining results and giving users a "good feeling," regardless of raw prediction accuracy C. The user community is too small for RMSE to be valid D. The researcher likely used the wrong Ground Truth
10	**Creating**	In a Constraint-based system, if no solution exists for a user's requirements (SRS), what is the purpose of calculating "Minimal Diagnoses"? A. To identify the most popular products as a fallback B. To debug errors in the product catalog's database C. To find the smallest set of user requirements to relax/remove to make a recommendation possible D. To automatically increase the user's budget constraint

Fuzzy Logic

No.	Bloom's Level	Question
1	Remembering	What is the term for a mathematical function used in Fuzzy Logic to define the degree of truth of a statement, mapping input values to the interval [0, 1]? A. Crisp Function B. Transfer Function **C. Membership Function** D. Probability Density Function
2	Remembering	The Mamdani-type Fuzzy Inference System typically uses which operator for the implication step (applying the rule strength to the output fuzzy set)? A. **Min (or product)** B. Max C. Sum D. Average
3	Understanding	Why is the Center of Gravity (Centroid) method generally considered the most robust defuzzification technique? A. It always selects the mean of the maximum output value. B. **It considers the entire shape and area of the aggregated fuzzy output set.** C. It only requires checking the peak value of the output fuzzy set. D. It avoids the need for a rule base entirely.
4	Understanding	Explain the consequence of using the Bounded Sum ($A \oplus B = \min(1, \mu A(x) + \mu B(x))$) as the S-norm (OR operator) compared to the standard Max operator. A. Bounded Sum always results in a lower membership degree. B. **Bounded Sum can allow the resulting membership degree to exceed Max(μA, μB), potentially reaching 1 faster when the two sets overlap significantly.** C. Bounded Sum is only used in Sugeno-type systems. D. Bounded Sum is equivalent to the Max operator in all cases.
5	Applying	A system has two rules contributing to the output 'Strong Wind': Rule 1 firing strength is 0.7, and Rule 2 firing strength is 0.9. If the output sets are aggregated using the standard Max (Union) operator, what is the height of the combined fuzzy set 'Strong Wind'? A. 0.7 B. **0.9** C. 1.6 D. 0.63
6	Applying	Given a fuzzy set A = { (1, 0.2), (2, 0.8), (3, 0.5) } and a fuzzy set B = { (1, 0.6), (2, 0.3), (3, 0.9) }. Calculate the membership values for the intersection $A \cap B$ using the standard Min operator. A. {(1, 0.6), (2, 0.8), (3, 0.9)} B. {(1, 0.8), (2, 1.1), (3, 1.4)} C. **{(1, 0.2), (2, 0.3), (3, 0.5)}** D. {(1, 0.4), (2, 0.5), (3, 0.4)}

7	Analyzing	In a complex control system, if the number of input variables and the number of linguistic terms per variable both increase, what is the primary consequence concerning the rule base? A. The defuzzification process becomes simpler. B. The system becomes less sensitive to input changes. C. **The rule base suffers from combinatorial explosion (exponential growth in the number of rules).** D. The Fuzzification step requires less computation.
8	Analyzing	Compare a Mamdani FIS that uses the Min operator for implication and the Center of Gravity (COG) for defuzzification versus a Sugeno FIS using weighted average. Which system is generally faster for real-time control, and why? A. Mamdani, because its output calculation is simpler. B. Mamdani, because it uses fuzzy sets directly. C. **Sugeno, because it avoids the computationally intensive integration required by the COG defuzzification method.** D. Sugeno, because it produces fuzzy outputs.
9	Evaluating	Evaluate the critical trade-off when selecting a narrow (steep slope) versus a wide (gentle slope) Membership Function for the linguistic term "High" in a temperature controller. A. A narrow MF increases the robust nature of the system. B. **A narrow MF increases the system's sensitivity to small input changes, potentially leading to oscillatory control.** C. A wide MF decreases the overlap between linguistic terms, simplifying the rule base. D. The shape of the MF only affects the aesthetic appearance of the control system.
10	Creating	Design a fuzzy rule that addresses the concept of hedging or uncertainty by linking a highly certain input state with a moderately conservative output action. Use the linguistic variables: Input_Certainty (Very_High, Moderate) and Output_Action (Aggressive, Cautious, Neutral). A. IF Input_Certainty IS Very_High THEN Output_Action IS Aggressive. B. IF Input_Certainty IS Moderate THEN Output_Action IS Neutral. C. **IF Input_Certainty IS Very_High THEN Output_Action IS Cautious.** D. IF Input_Certainty IS Low THEN Output_Action IS Aggressive.

Linear Regression

No.	Bloom's Level	Question
1	Remembering	What is the primary goal of Simple Linear Regression? A. To find the best fitting curve. **B. To model the relationship between one independent variable (X) and one dependent variable (Y) using a straight line.** C. To classify data points into discrete categories. D. To minimize the sum of absolute errors.

2	Remembering	In the linear regression equation Y= $\beta_0 + \beta_1 X + \epsilon$ represent? A. The Y-intercept. **B. The slope of the regression line.** C. The residual error. D. The expected value of Y when X is zero.
3	Understanding	What is the fundamental difference between Simple Linear Regression and Multiple Linear Regression? A. Simple Linear Regression has continuous variables; Multiple Linear Regression has categorical variables. **B. Simple Linear Regression uses one predictor variable; Multiple Linear Regression uses two or more predictor variables.** C. Multiple Linear Regression is only used for non-linear relationships. D. Simple Linear Regression cannot handle noise (ϵ).
4	Understanding	Explain the meaning of a coefficient of determination (R^2) value of 0.85 in a model. A. 85% of the data points fall directly on the regression line. **B. 85% of the variance in the dependent variable (Y) is predictable from the independent variable (X).** C. The model has an 85% chance of predicting the correct value. D. The correlation between X and Y is 0.85.
5	Applying	A model predicts house price (Y) based on square footage (X) with the equation Price = 50,000 + 100 · SqFt. If a house is 2,000 SqFt, what is the predicted price? **A. $250,000** B. $200,000 C. $150,000 D. $100,000
6	Applying	Which loss function is most commonly minimized during the training of a standard Linear Regression model using the Ordinary Least Squares (OLS) method? A. Mean Absolute Error (MAE) B. Binary Cross-Entropy **C. Mean Squared Error (MSE)** D. Log Loss
7	Analyzing	When performing residual analysis, if you observe a distinct pattern (e.g., a parabolic shape) in the plot of residuals versus fitted values, what does this indicate about the linear regression model? A. The model is highly accurate. **B. The assumption of linearity has been violated.** C. The data is perfectly correlated. D. The residuals are normally distributed.
8	Analyzing	Why is it crucial to check for multicollinearity in Multiple Linear Regression, and what statistical measure is often used to detect it? A. Multicollinearity makes the model faster; VIF is used. **B. Multicollinearity inflates the standard errors of the coefficients, making them unstable; Variance Inflation Factor (VIF) is used.** C. Multicollinearity is only a problem in simple regression; R-squared is used. D. Multicollinearity implies a perfect fit; ANOVA is used.

9	Evaluating	Evaluate the trade-off between using a high-degree polynomial regression model (which perfectly fits the training data) versus a simple linear regression model for future prediction. Which model is generally preferred for generalization, and why? **A. The simple linear regression model, because the polynomial model is prone to overfitting and will perform poorly on new, unseen data.** B. The polynomial model, because a perfect fit on the training data guarantees high accuracy everywhere. C. The choice depends only on the R-squared value, not on generalization. D. Both models generalize poorly; only non-parametric models should be used.
10	Creating	Propose a hypothetical feature transformation (e.g., squaring, logging) for the independent variable X in a dataset where the relationship between X and Y is initially exponential, and briefly explain why this transformation is necessary for Linear Regression. **A. Transform X to log(X). This is necessary because Linear Regression assumes a linear relationship, and the logarithmic transformation effectively linearizes the initial exponential relationship.** B. Transform X to X^2. This is necessary to satisfy the normality assumption of the errors. C. Transform Y to Y^{-1}. This is necessary to deal with heteroscedasticity. D. No transformation is necessary; Linear Regression can handle exponential relationships directly.

Logistic Regression

POST-TEST - LOGISTIC REGRESSION

No.	Bloom's Level	Question
1	Remembering	What is the primary difference between Linear Regression and Logistic Regression regarding the dependent variable (Y)? A. **Linear Regression predicts continuous values; Logistic Regression predicts probabilities for binary outcomes.** B. Linear Regression uses categorical inputs; Logistic Regression uses continuous inputs. C. They both predict continuous values but use different optimization methods. D. Logistic Regression cannot handle multiple inputs.
2	Remembering	Which function is applied in Logistic Regression to map the output of the linear equation into a probability range between 0 and 1? A. Rectified Linear Unit (ReLU) B. Softmax Function **C. Sigmoid Function** D. Tanh Function

3	Understanding	Why does Logistic Regression use the Sigmoid function instead of directly using the linear equation Y= $\beta_0 + \beta_1 X + \epsilon$? A. To ensure the predicted outcome is non-negative. **B. To transform the output into a value that can be interpreted as a valid probability (between 0 and 1).** C. To minimize the Mean Squared Error (MSE). D. To correct for multicollinearity.
4	Understanding	In Logistic Regression, the log-odds (logit) is defined as the logarithm of the ratio of the probability of success (p) to the probability of failure ($1 - p$). What is the formula for the log-odds? **A.** $\ln(\frac{p}{1-p})$ B. p + (1 - p) C. 1/p D. e^p
5	Applying	A Logistic Regression model provides a log-odds (logit) value of 0 for a specific customer. What is the calculated probability (p) that this customer belongs to the positive class? A. 0 **B. 0.5** C. 1 D. 0.731
6	Applying	Which mathematical function is typically minimized as the cost function (loss function) when training a Logistic Regression model? A. Mean Squared Error (MSE) B. R-squared **C. Cross-Entropy Loss (or Log Loss)** D. Sum of Absolute Errors (SAE)
7	Analyzing	How should a coefficient β be interpreted in Logistic Regression? A. A one-unit increase in X increases the predicted probability p by β. B. A one-unit increase in X increases the predicted Y by β. **C. A one-unit increase in X increases the log-odds of the positive class by β.** D. A one-unit increase in X decreases the residual error by β.
8	Analyzing	In a classification dataset, if the data points are perfectly separated by a linear boundary, why might training a standard (unregularized) Logistic Regression model lead to numerical issues? A. The model will converge immediately. **B. The coefficients (β) will approach infinity, indicating complete separation (perfect fit).** C. The Cross-Entropy Loss will become zero, halting the training process prematurely. D. The Sigmoid function will output only 0s.

9	Evaluating	For a system designed to predict whether a manufacturing machine will fail (positive class) or run normally (negative class), which metric should be prioritized and why? A. Precision, because minimizing False Positives is crucial for cost savings. **B. Recall, because minimizing False Negatives (missed failures) is essential to prevent costly operational damage.** C. Accuracy, because the overall correctness matters most. D. F1 Score, because it balances Precision and Recall equally.
10	Creating	Propose a specific architecture or method to adapt a Binary Logistic Regression model for a problem with three or more mutually exclusive classes (e.g., predicting Low, Medium, or High risk levels). A. Use the Kernel Trick to map data to a higher dimension. B. Use Simple Linear Regression for each class independently. **C. Use Multinomial Logistic Regression (Softmax Regression) or the One-vs-Rest (OvR) approach.** D. Use the Sigmoid function multiple times on the same output.

Latent Dirichlet Allocation

No.	Bloom's Level	Question
1	Remembering	In Latent Dirichlet Allocation (LDA), what does each topic represent? A. A single document B. A probability distribution over words C. A cluster of documents D. A label assigned manually
2	Remembering	What is the role of the Dirichlet distribution in LDA? A. It defines distances between documents B. It generates Gaussian noise C. It acts as a prior over topic distributions D. It normalizes word counts
3	Understanding	Why is LDA considered a generative probabilistic model? A. It classifies documents into categories B. It describes how documents could be generated from latent topics C. It directly predicts labels D. It minimizes prediction error
4	Understanding	What is the key assumption LDA makes about documents? A. Each document belongs to exactly one topic B. Words are independent of topics C. Each document is a mixture of topics D. Topics are fixed and predefined

5	Applying	If a document has topic distribution θ = [0.7 Sports, 0.3 Politics], what does this imply? A. Document belongs only to Sports B. 70% of words are generated from Sports topic C. Document is labeled Sports D. Topics are deterministic
6	Applying	During Gibbs Sampling in LDA, what is iteratively updated? A. Document labels B. Topic assignments for each word C. Vocabulary size D. Neural weights
7	Analyzing	What happens if the number of topics K is too large? A. Topics become more general B. Topics may become fragmented or incoherent C. Model stops training D. Documents cannot be represented
8	Analyzing	A company uses LDA to infer user interests from private messages without explicit consent. What is the primary ethical concern? A. Overfitting B. Privacy violation and lack of informed consent C. Low topic coherence D. High computational cost
9	Evaluating	When deploying LDA for user profiling in a recommendation system, which factor is most critical from an ethical standpoint? A. Maximizing number of topics B. Minimizing perplexity C. Ensuring transparency and user consent D. Increasing vocabulary size
10	Creating	How would you modify an LDA-based system to address privacy concerns? A. Increase number of topics B. Use anonymization or differential privacy techniques C. Remove stopwords D. Reduce dataset size

Deep Neural Networks

No.	Bloom's Level	Question
1	Remembering	What defines a deep neural network? A. Large dataset B. Multiple hidden layers C. High accuracy D. Large input size
2	Remembering	Which activation function is widely used in modern DNNs? A. Sigmoid B. ReLU C. Tanh D. Step
3	Understanding	Why are non-linear activation functions important? A. Reduce training time B. Enable learning complex patterns C. Normalize data D. Prevent overfitting
4	Understanding	What is the purpose of backpropagation? A. Initialize weights B. Update weights using gradients C. Normalize inputs D. Reduce dataset size
5	Applying	What is computed before activation in a neuron? A. x1 + x2 B. w1 + w2 C. Weighted sum + bias D. x1x2
6	Applying	What happens if learning rate is too high? A. Slow training B. Better accuracy C. Divergence of loss D. No effect
7	Analyzing	What is the vanishing gradient problem? A. Gradients explode B. Gradients approach zero in deep layers C. Loss becomes zero D. Weights freeze completely

8	Analyzing	A DNN used for hiring decisions shows lower acceptance rates for a specific demographic group. What is the likely issue? A. Overfitting B. Data leakage C. Algorithmic bias in training data D. High variance
9	Evaluating	A highly accurate DNN cannot explain its decisions in a medical diagnosis system. What is the main concern? A. Model size B. Training speed C. Lack of interpretability and trust D. Low precision
10	Creating	How would you improve fairness in a DNN used for loan approval? A. Increase layers B. Remove sensitive attributes and apply fairness-aware training C. Increase learning rate D. Use larger dataset only

Word Embedding

No.	Bloom's Level	Question
1	Remembering	What is the purpose of Word Embedding? A. One-hot encoding B. Dense vector representation of words C. Count word frequency D. Grammar parsing
2	Remembering	Which model is commonly used to learn embeddings? A. K-means B. Word2Vec C. Apriori D. PCA
3	Understanding	Why are embeddings better than one-hot vectors? A. Faster computation B. Capture semantic similarity C. No training needed D. Smaller vocabulary

4	Understanding	Difference between CBOW and Skip-gram? A. CBOW predicts context B. CBOW predicts word from context; Skip-gram predicts context from word C. Same model D. Skip-gram uses no NN
5	Applying	"king - man + woman ≈ ?" A. prince B. queen C. girl D. mother
6	Applying	If two words share similar contexts, what happens? A. Orthogonal vectors B. Similar embeddings C. Removed words D. Identical tokens
7	Analyzing	What is a limitation of Word2Vec? A. Sparse vectors B. Cannot capture multiple meanings (polysemy) C. Needs labels D. No training
8	Analyzing	A Word2Vec model learns associations like "doctor → male" and "nurse → female". What does this indicate? A. Model accuracy B. Learned societal bias from training data C. Overfitting D. Data noise
9	Evaluating	Why is bias in word embeddings problematic in real-world applications? A. Slower training B. Larger vectors C. It can propagate unfair or discriminatory decisions in downstream systems D. Reduced vocabulary
10	Creating	How would you mitigate bias in word embeddings? A. Increase vector size B. Apply debiasing techniques or retrain on balanced data C. Remove rare words D. Use one-hot encoding

SECTION 2:

CASE STUDY: CORAL REEF MONITORING AND RESTORATION SYSTEM (PROJECT CORALGUARD)

Background: Project CoralGuard is an AI-driven ecosystem designed to preserve and restore coral reefs severely degraded by climate change. The system coordinates a network of Autonomous Underwater Vehicles (AUVs) and fixed undersea sensor stations to perform 24/7 monitoring and precision biological interventions.

Primary Data Sources:

1. **Maintenance Logs:** A database recording the specific microbes, minerals, and nutrients injected into coral clusters during each robot mission (Transaction-based data).
2. **IoT Sensor Grid:** Continuous recording of environmental parameters such as pH levels, seawater temperature, dissolved oxygen concentration, and undersea light intensity.
3. **Field Reports & Research Journals:** Qualitative professional notes written in natural language by marine biologists regarding coral health, disease symptoms, and symbiotic species diversity.
4. **Vision Stream:** High-resolution video and image data from underwater cameras used to track coral morphological changes and classify various fish species within the protected area.

Learner's Task: Based on the context and data sources of Project CoralGuard, you are required to apply the algorithms and models from your 02 selected topics to analyze and solve the specific technical scenarios provided in the following section.

Association Rules Mining

Question 1: In the Maintenance Logs database of CoralGuard, each "Transaction" records the components pumped into the coral reef by the robot. If an engineer wants to find groups of microbes frequently used together, what does an Itemset represent in this problem?

A. A list of coordinate locations of the coral reefs.
B. A collection of specific microbes and nutrients included in a single maintenance mission.
C. The total time the autonomous underwater vehicle (AUV) spent on missions.
D. The number of coral images collected from the Vision Stream.

Question 2: The research team found an association rule: {Microbe A} => {Mineral B} with a confidence Conf = 80%. Based on the confidence formula, which of the following conclusions is correct?

A. Mineral B appears in 80% of the total maintenance missions of the entire project.
B. In 80% of the times the robot pumped Microbe A, Mineral B was also pumped at the same time.
C. Microbe A and Mineral B only appear together in exactly 80 transactions.
D. The Support of the itemset {Microbe A, Mineral B} is definitely 80%.

Question 3: While running the Apriori algorithm, the system determines that the itemset {Nutrient X, Microbe Y} does not meet the minSup (minimum support) threshold. According to the Apriori property, how should the robot handle the candidate itemset {Nutrient X, Microbe Y, Mineral Z}?

A. Continue scanning the database to calculate the support count for this three-item set.
B. Automatically consider it a frequent itemset because of the addition of Mineral Z.
C. Prune it immediately because any superset of an infrequent itemset must also be infrequent.
D. Increase the minSup threshold to keep this itemset for future analysis.

Question 4:

The CoralGuard system proposes two different rules to restore diseased corals:

1. {Nutrient Alpha} => {Recovery}: Sup = 2%, Conf = 90%
2. {Nutrient Beta} => {Recovery}: Sup = 15%, Conf = 40% If the research station needs a solution that ensures the highest reliability for rare coral species, even if the frequency of application is low, which rule should be prioritized?

A. Rule 2, because it has higher support, proving it is effective for a wider range of cases.
B. Both rules because they share the same consequent ("Recovery").
C. Rule 1, because the high confidence indicates a nearly certain recovery when Alpha is used.
D. Neither rule, because the support values are too low for a large-scale intervention.

Question 5: To detect environmental risks early, a scientist wants to find rules that lead to the condition {Coral Bleaching}. How would you set up the association rules mining task based on the learned sub-problems?

A. Only scan transactions that do not contain any nutrients or microbes.
B. Fix the consequent (Y) of the rule as the itemset {Coral Bleaching} and find antecedents (X) with high confidence leading to this state.
C. Search only for rules with the lowest possible support because disasters are rare events.
D. Use an exhaustive search to find every possible rule combination without applying any threshold conditions.

Recommender System

Question 1: The CoralGuard system needs to suggest specific intervention sites to its fleet of AUVs (Autonomous Underwater Vehicles). If we treat this as a recommendation problem, what would be the most logical way to define the User-Item relationship?

A. The Users are the fish species, and the Items are the coral nutrients.
B. The Users are the specific undersea stations (locations), and the Items are the possible intervention techniques (e.g., pH balancing, mineral injection).
C. The Users are the human biologists, and the Items are the high-resolution videos from the Vision Stream.
D. The Users are the AUV batteries, and the Items are the charging docks.

Question 2: A scientist wants to implement a Content-based Filtering approach to suggest new rescue locations for a robot. Which of the following data points would be the primary focus for this method?

A. The specific features of the reef, such as depth, coral species type, and local water temperature.
B. The historical ratings given by other robots that have visited similar reefs in the past.
C. The total number of maintenance logs generated by the entire project over the last year.
D. A random selection of reefs to ensure the robot explores new areas of the ocean.

Question 3: A User-based Collaborative Filtering algorithm is used to suggest restoration strategies for a reef cluster named Reef-Alpha. The system identifies that Reef-Beta and Reef-Gamma have very similar environmental profiles and past successful outcomes to Reef-Alpha. How will the system generate a recommendation?

A. By analyzing the genetic code of the coral in Reef-Alpha only.
B. By suggesting a strategy that has never been used before to test its effectiveness.
C. By recommending strategies that were successful in Reef-Beta and Reef-Gamma but have not yet been tried in Reef-Alpha.
D. By calculating the linear regression of the temperature changes in all three reefs.

Question 4: Project CoralGuard recently added a brand-new sensor station in a remote, unexplored area of the ocean. The system is struggling to provide recommendations for this station because there is no historical intervention data yet. This is an example of which common Recommender System challenge?

A. Overfitting.
B. Data Sparsity.
C. Cold Start Problem.
D. Filter Bubble.

Question 5: You are designing a system to recommend "Emergency Rescue" sites for robots when a sudden heatwave occurs. You have two models:

- **Model A:** High Precision (When it recommends a site, it is definitely a high-risk emergency).
- **Model B:** High Recall (It finds every single emergency site, but also includes some healthy sites by mistake). If the robot's battery life is extremely limited and it can only visit 2 sites per day, which model should you prioritize for the recommendation engine?

A. Model A, because it ensures the robot does not waste its limited battery on healthy sites.
B. Model B, because it is better to visit every site even if some are not in danger.
C. Neither, because recommendation systems are not suitable for emergency situations.
D. A random selection model to ensure the battery is used evenly across the ocean.

Fuzzy Logic

Question 1: A marine biologist states that "Seawater Temperature is Warm" when the temperature is 28.5°C. If the fuzzy set "Warm" has a triangular membership function peaking at 29°C (μ=1), starting at 27°C (μ=0), and ending at 30°C (μ=0), what is the degree of membership for 28.5°C?

A. 0.5
B. 0.75
C. 0.6
D. 0.9

Question 2: A Fuzzy Inference System (FIS) is designed to determine Intervention_Urgency based on two inputs: pH_Level and Dissolved_Oxygen. If the rule base uses the Mamdani method, why is it necessary to convert the final aggregated fuzzy output (e.g., 'High Urgency') into a single crisp value before the AUV can act?

A. To minimize the cost function.
B. To prevent the rule base from expanding.
C. **Because the AUV requires a definite, crisp control signal (e.g., a specific motor speed or dosage amount).**
D. To calculate the R-squared value of the model.

Question 3: The Field Reports contain qualitative terms like "Coral Health is Excellent" and "Light Intensity is Very Low." How does the Fuzzifier component of the CoralGuard system handle the precise sensor reading of 450 lux to activate the linguistic term "Very Low"?

A. It assigns a binary 0 or 1 value based on a fixed threshold.
B. It converts the 450 lux into a log-odds ratio.
C. It calculates the degree of membership (μ) of 450 lux in the predefined fuzzy set for "Very Low Light Intensity."
D. It calculates the mean absolute error.

Question 4: A control engineer proposes using a crisp logic system (fixed thresholds) to regulate the High CO2 injection. If the threshold for "High CO2" is set exactly at 8.0 ppm, and the sensor reading fluctuates rapidly between 7.9 ppm and 8.1 ppm, evaluate the likely impact on the AUV's control valve compared to a Fuzzy Logic system.

A. The Fuzzy Logic system would be more complex and slower.
B. The crisp system would cause abrupt, oscillatory, and continuous opening/closing of the valve, whereas the Fuzzy system would allow for smooth, proportional control changes.
C. The crisp system would perfectly handle the noise.
D. Both systems would behave identically.

Question 5: The system needs a rule to prevent coral damage. Propose a single Mamdani-type fuzzy rule that dictates a maximum intervention response using inputs Seawater Temperature (Cool, Hot) and pH Level (Acidic, Neutral).

A. IF Temp IS Hot OR pH IS Neutral THEN Intervention IS Max.

B. IF Temp IS Cool THEN Intervention IS Min.

C. IF Seawater Temperature IS Hot AND pH Level IS Acidic THEN Intervention Output IS Maximum (Urgent).

D. IF Temp IS Hot AND pH IS Neutral THEN Intervention IS Medium.

Linear Regression

Question 1: Scientists want to model the linear relationship between Seawater Temperature (X) and Coral Growth Rate (Y). If the calculated regression equation is Growth Rate = 0.5 - 0.15 · Temp, what is the predicted Growth Rate if the Seawater Temperature is 30◦C?

A. 0.5 units/day

B. 30 units/day

C. -4.0 units/day

D. 4.0 units/day

Question 2: A Multiple Linear Regression model is built using data from the IoT Sensor Grid to predict Coral Health Score (Y) based on Temperature X1 and pH X2. If the coefficient for Temperature X1 is β1 = -0.8, how should this value be interpreted?

A. A 1-unit increase in Temp increases Coral Health Score by 0.8 units, holding pH constant.

B. A 1-unit increase in Temp decreases the log-odds of health by 0.8.

C. A 1-unit increase in Seawater Temperature decreases the Coral Health Score by 0.8 units, assuming the pH level remains constant.

D. Temperature is perfectly correlated with Coral Health.

Question 3: A Linear Regression model is used to predict the growth rate of specific coral species. A scatter plot of the residuals versus the predicted growth rate shows a distinct fanning-out pattern (residuals get wider as the predicted value increases). Which key assumption of Linear Regression is most likely being violated?

A. Independence of observations

B. Normality of residuals

C. Homoscedasticity (constant variance of residuals)

D. Linearity of the relationship

Question 4: The research team has continuous temperature data X_1 and dissolved oxygen X_2 and wants to predict coral bleaching risk (Y, a continuous score from 0 to 100). The current model has an R^2 of 0.95, but the biologists suspect the relationship is not perfectly linear. Why is it important to use residual analysis (plotting residuals vs. X_1 or X_2 despite the very high R^2 value?

A. High R^2 always means the model is perfect.

B. High R^2 does not guarantee linearity; a structured pattern in the residuals could reveal a non-linear relationship (e.g., quadratic) that requires transformation.

C. Residual analysis is only for classification models.

D. R^2 is only useful for binary data.

Question 5: The relationship between undersea light intensity (X) and photosynthetic rate (Y) of the coral displays a classic non-linear, diminishing returns curve (the rate increases quickly at first, then slows down). Propose a transformation for the independent variable X that would enable the use of Multiple Linear Regression to model this relationship effectively.

A. Transform Y to 1/Y

B. Transform X to X^2

C. Transform X to log(X)

D. Transform Y to log(Y)

Logistic Regression

Question 1: The CoralGuard system uses Logistic Regression to predict the probability of a reef cluster exhibiting "Bleaching Symptoms"(Y = 1) based on Seawater Temperature and pH levels. Why is Logistic Regression, using the Sigmoid function, the appropriate choice over Linear Regression for this task?
A. Logistic Regression is faster to train on the IoT Sensor Grid data.
B. The Sigmoid function ensures that the predicted outcome, the risk of bleaching, is scaled to a meaningful probability value between 0 and 1.
C. Linear Regression cannot handle continuous inputs like Temperature.
D. Logistic Regression automatically handles categorical variables without encoding.
Question 2: A Logistic Regression model is trained to predict "Disease Outbreak" (Y = 1). The linear combination of inputs results in a log-odds (logit) value of 2.0 for a specific sensor station. Calculate the approximate probability (p) of a disease outbreak at this station.
A. 0.2
B. 0.88
C. 0.5
D. 1.0
Question 3: The model finds a negative coefficient (β) of -2.5 for the independent variable Dissolved Oxygen Concentration when predicting the probability of "Coral Mortality" (Y=1). How should the marine biologist interpret this finding?
A. High Dissolved Oxygen is strongly associated with increased Coral Mortality.
B. A one-unit increase in Dissolved Oxygen decreases the log-odds of Coral Mortality, suggesting it reduces the risk.
C. Dissolved Oxygen must be a categorical variable.
D. The probability of Coral Mortality is 2.5.
Question 4: The AUV fleet is tasked with intervention based on the predicted probability of "Reef Collapse" (Y=1). Missing a collapsing reef (False Negative) is catastrophic. The model needs a classification threshold. Which evaluation metric should the team prioritize to minimize this catastrophic error?
A. Precision (minimizing False Positives)
B. Recall (minimizing False Negatives)
C. F1 Score (balance)
D. Accuracy (overall correctness)
Question 5: A researcher wants to build a Multinomial Logistic Regression model to classify the severity of coral stress into three mutually exclusive categories: {Low Stress, Moderate Stress, Severe Stress}. Which approach must they use to adapt the binary Logistic Regression framework to handle these three classes?
A. Train separate Linear Regression models for each class.
B. Use Multinomial Logistic Regression (Softmax function) or the One-vs-Rest (OvR) strategy.
C. Use the Sigmoid function three times independently.
D. Combine all three categories into a single binary outcome.

Latent Dirichlet Allocation

Question 1: The CoralGuard system uses Field Reports & Research Journals (text data) to extract latent topics about coral health. In this context, what would a "topic" most likely represent?

A. A specific coral reef location
B. A cluster of documents written by one biologist
C. A distribution of words describing a biological theme (e.g., disease, symbiosis, bleaching)
D. A single keyword like "coral"

Question 2: A document (field report) has the topic distribution: [0.6 Disease, 0.4 Symbiosis]. How should this be interpreted?

A. The document only discusses disease
B. 60% of the words are generated from the Disease topic
C. The document is labeled as Disease
D. Topics are assigned randomly

Question 3: The research team sets the number of topics K too low (e.g., K=2). What is the most likely consequence?

A. Topics become too specific
B. Topics become overly broad and mix unrelated concepts
C. Model fails to run
D. Vocabulary size decreases

Question 4: The system uses LDA to automatically profile research notes written by individual marine biologists to evaluate their performance without informing them. What is the primary concern?

A. Model underfitting
B. Privacy and lack of informed consent
C. Low perplexity
D. Topic redundancy

Question 5: You want to track how coral ecosystem themes evolve over time (e.g., increase in bleaching-related topics). Which extension of LDA is most appropriate?

A. K-means clustering
B. Dynamic Topic Model (DTM)
C. Linear Regression
D. Apriori Algorithm

Deep Neural Networks

Question 1: The Vision Stream provides underwater images for coral classification. Why is a Deep Neural Network suitable for this task?

A. It reduces dataset size
B. It can automatically learn hierarchical visual features
C. It requires no training
D. It only works with text

Question 2: A CNN model is trained to detect coral bleaching from images. What is the role of convolutional layers?

A. Store data
B. Extract spatial features like edges and textures
C. Normalize outputs
D. Reduce dataset size

Question 3: During training, the model performs very well on training data but poorly on new reef images. What is the issue?

A. Underfitting
B. Overfitting
C. Data normalization
D. Gradient explosion

Question 4: The DNN model consistently misclassifies coral health in reefs located in deeper waters because such data was underrepresented in training. What is the issue?

A. High variance
B. Activation function error
C. Overfitting
D. Algorithmic bias due to imbalanced data

Question 5: The CoralGuard system must select a model for coral disease diagnosis.

- Model A: Accuracy = 92%, Explainable (can show feature importance to biologists)
- Model B: Accuracy = 96%, Black-box DNN (no interpretability)

In this high-stakes ecological intervention, where incorrect actions may damage fragile coral ecosystems, what is the most appropriate strategy?

A. Always deploy Model B because higher accuracy guarantees better outcomes
B. Deploy Model A because interpretability allows experts to validate and override decisions when necessary
C. Randomly switch between both models to balance performance
D. Avoid deploying any model due to risk

Word Embedding

Question 1: The system processes Field Reports using Word Embeddings. What is the main advantage in this context?

A. Reduce text length
B. Capture semantic similarity between biological terms
C. Remove stopwords
D. Count word frequency

Question 2: If the embedding model learns that "bleaching" is close to "temperature stress" in vector space, what does this indicate?

A. Words are unrelated
B. They frequently co-occur in similar contexts
C. They are identical
D. Model failed training

Question 3: Why might standard Word2Vec struggle when analyzing coral-related scientific text?

A. Too many words
B. Cannot handle images
C. Cannot distinguish multiple meanings of domain-specific terms
D. Requires labels

Question 4: The embedding model is trained on historical research that under-represents certain coral species. What is a potential risk?

A. Faster training
B. Bias in representing less-studied species
C. Smaller vocabulary
D. Higher accuracy

Question 5: You want embeddings that understand context differences between sentences like:

- "coral bleaching event"
- "data bleaching process"

What is the best approach?

A. One-hot encoding
B. Use contextual embeddings (e.g., transformer-based models)
C. Remove ambiguous words
D. Use TF-IDF only

SECTION 3:

Instructions: Please rate your level of agreement with the following statements on a scale from 1 to 5.

(1: Strongly Disagree | 2: Disagree | 3: Neutral | 4: Agree | 5: Strongly Agree)

Part 1: Perceived Usefulness

1. AI tools help me clarify complex technical concepts more effectively.
2. AI is a valuable assistant for summarizing and connecting different parts of the lesson.

Part 2: Confidence

3. I feel more capable of solving technical exercises when I have AI support.
4. I am confident in my ability to explore new technical topics with AI as a guide.

Part 3: Critical Thinking

5. I proactively verify AI-generated information against official textbooks or slides.
6. I can identify if an AI's technical explanation is illogical or incorrect.

Part 4: Dependency

7. I use AI to get hints or ideas rather than just asking for the final answer.
8. I prioritize my own logical reasoning over the solutions suggested by AI.

Part 5: Trust & Readiness

9. I trust AI's support, but I take full responsibility for the accuracy of my work.
10. I am ready to apply this AI-supported learning style to my future studies.